

The Politics of Technology (lecture note)

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Introduction

In the last lecture on ethics, you learnt that there are different ways for determining the goods and bads of technology, and that there is no objective way for evaluating a given technology. However, not only ethicists debate technology, we all do. In fact, emerging technologies are often quite controversial because different actors have differing perceptions of and interests in these technologies. For example, Sam Altman, the CEO of OpenAI, would love to train his large language models (LLMs) on the Internet for free. Artists and other content creators vehemently disagree that this is a good idea, because it threatens their livelihood. As students, you might think that ChatGPT is a very useful tool to write an essay. As a teacher, I might worry about what that does to learning, and whether it helps or prevents you to learn what matters when writing an essay. So, the question of whether people like or dislike a given technology is not only a question of ethics, it is a question of **politics**, that is, a struggle of interests and values that are rooted in personal experience, political beliefs, and social positions.

This lecture will teach you about the politics of technology, and how, as a result, it can become controversial. In particular, we will put into question **three myths** often presented in debates about technology: its autonomy, value-neutrality, and inherent goodness. To challenge these beliefs, the lecture introduces three corresponding concepts: **interpretative flexibility**, **script**, and **political technologies**. These concepts help you understand and analyse how technology's impact on society is mediated through users' concerns and perceptions, how engineers inscribe assumptions about these users into their designs and, in turn, how this constrains how we live and organise society.

Technology is interpretatively flexible

A common belief is that technology is autonomous. It manifests in many different forms: technology is portrayed as "inevitable", "out of control" or "a force of nature". This frames the relationship between technology and society as a one-way street and beyond human influence. As a result, proponents of this position find attempts at steering technological development a naïve and vain endeavour.

However, from research we know that technology is not separate from society. Sociologists since the 1970s have shown that technology does not only determine human action but that human action also determines technology. These researchers looked at what actually happened when new technologies get diffused in society, that is, when people started using them. They show that technologies can have radically different meanings for different groups of users, who may come up with entirely new and unforeseen ways of using a given technology.

Example: In 1920s–30s rural America, farmers repurposed trucks to power stationary machinery, going beyond their intended use, i.e., transportation. Their adaptations led manufacturers to add features like power take-offs. This illustrates interpretative flexibility: users redefine a technology's meaning and influence its design, challenging the idea of fixed,

designer-driven technological development (for more details, see the text by Kline and Pinch in the Further Readings section).

So, technology can exhibit what Ronald Kline, Trevor Pinch and others call **interpretative flexibility**. This concept describes a state when different interpretations of what a technology means or does co-exist or conflict. These differences stem from social factors, for example, differences in culture, gender, social position etc., and they can appear both at the design and use stage of a technology. This is important to understand controversies, because it reminds us that disagreements about technology are not just about good and bad consequences. Rather, they go deeper in that they stem from fundamentally different interpretations of what a technology 'is' in relation to the values that different social groups hold.

Interpretative flexibility is usually not a permanent state. Rather, disagreements are settled over time. This process is called **closure**. Closure is the result of interaction between users and designers. The stability of technology is thus not only dependent on a technology's inherent material properties but due to overarching values and norms people hold and attach to technology.

Technology is not neutral

Another common belief when discussing whether a given technology is good or bad, is that technology itself is neutral. What counts is only how people use it. Proponents of this position will say that, for example, a gun does not shoot by itself, a person does. Social media is not inherently bad, certain people just have predispositions that make them addicted to it. We encounter this argument especially when talking about the possible negative consequences of a technology. For example, when saying that the 'risk' of a given technology only stems from potential bad actors "misusing" it. In the end, what counts for this position is not technology itself but only people's intentions.

However, from research we know that this is not the whole story. Technologies are not neutral, passive objects. Rather, their design may prescribe certain ways of using it and prevent others. Yes, a gun needs a human being to be lethal, but placing a gun in a room of people will certainly change the atmosphere and dramatically increase the likelihood for something bad to happen. In the case of social media we know that platforms like TikTok or Instagram are designed to keep you engaged and thus play a crucial role in problematic internet use.

In sociology, we say that technologies have a **script**, that is,

"when technologists define the characteristics of their objects, they necessarily make hypotheses about the entities that make up the world into which the object is to be inserted. Designers thus define actors with specific tastes, competences, motives, aspirations, political prejudices, and the rest, and they assume that morality, technology, science, and economy will evolve in particular ways. A large part of the work of innovators is that of 'inscribing' this vision of (or prediction about) the world in the technical content of the new object. I will call the end product of this work a 'script' ..." (Akrich 1992: 207f.)

This process of inscription can take many forms. It may be through coding an algorithm, designing a user interface, or building a chair. In all these cases, designers embed assumptions and intended uses into technologies, producing scripts that guide user behaviour. However, this does not mean that these scripts are always realised in practice. As

we have learnt above, users may follow, resist, or reinterpret these scripts, something that Madeleine Akrich calls “de-description”. In fact, sociological studies of technology often focus on cases where scripts are resisted or reinterpreted to show that technology is not neutral and that the scripts designers inscribe into technology are, to an extent, malleable. The bottom line is that the degree to which technology is interpretatively flexible crucially depends on how rigidly technological objects are scripted.

Example: Akrich describes a case where a lighting kit for off-grid use in Africa included a battery-powered lamp. Designers scripted users to recharge batteries at a central location, assuming collective responsibility. However, users altered the system for individual use, revealing how embedded scripts can be resisted, reinterpreted, or fail entirely (for a more detailed description, see the text by Akrich in the Further Readings section)

Technology is political

Finally, there is a common belief, especially in policy, engineering, and the private sector, that technology ultimately leads to positive social outcomes and progress, and that these outcomes are inherently beneficial, a “net good” for society. Proponents of this position argue that technology allows us to solve almost any problem, be it to cure cancer or address climate change. A very recent example is the discourse on generative AI, which is hailed as a allrounder fix to almost any challenge we currently face.

Here, scholars in political science counter that instead of being inherently good technologies have specific **political characteristics**, intentionally or unintentionally influencing social order for good or for bad. Hence, their goodness depends on the design of those political properties as well as, of course, how these properties are perceived by users and how they affect them. A prominent proponent of this political view of technology, Langdon Winner, argues in his seminal paper (see Further Readings section) that rather than treating technology as a solution to all of our problems, we should view them more as laws, as attempts to influence how society is run.

He says that some technologies are **political technologies**, of which he distinguishes two types:

- **Explicitly political technology:** Designed deliberately to support certain power relations or control, that is, to benefit some and exclude others.
- **Inherently political technology:** Technologies requiring or promoting particular governance structures, that is, ways of organising social order.

Example for an explicitly political technology: The mechanized tomato harvester, developed with university and industry support, displaced thousands of farmworkers and favoured large-scale agribusiness. Its design and adoption were politically charged, reinforcing economic inequality and undermining small farms—demonstrating how technology can be deliberately shaped to serve the interests of certain groups – in this case of big agricultural corporations.

Example for an inherently political technology: Nuclear power requires centralized control, technical expertise, and strict security—favouring hierarchical, authoritarian structures. Its technical requirements make certain political arrangements more viable than others, regardless of intention. Thus, it exemplifies how some technologies inherently necessitate specific forms of social and political organization.

Conclusion

Responsible engineers should not just assume that technologies are neutral, autonomous, or inherently good. They should engage with diverse users early, reflect on the values and assumptions embedded in designs, and consider the broader social and political effects of their work. In doing so, engineers can better anticipate unintended consequences and foster inclusive, responsible innovation that aligns with societal needs rather than reinforcing existing inequalities or blindly chasing progress for its own sake.

Further readings

For a general introduction into the topic:

Feng P (2000) Rethinking technology, revitalizing ethics: Overcoming barriers to ethical design. *Science and Engineering Ethics* 6(2): 207–220.

For an introduction into the specific concepts discussed above:

For the concept of script: Akrich M (1992) The De-Scriptio of Technical Objects. In: Bijker WE and Law J (eds) *Shaping Technology/Building Society: Studies in Sociotechnical Change*. Inside technology. Cambridge, MA: MIT Press, pp. 205–224.

For the concept of interpretative flexibility: Kline R and Pinch T (1996) Users as Agents of Technological Change: The Social Construction of the Automobile in the Rural United States. *Technology and Culture* 37(4): 763–795.

For the concept of political technologies: Winner L (1980) Do Artifacts Have Politics? *Daedalus* 109(1): 121–136.